Lesson Plan: Artificial Lighting – LED Color Growth Experiment



Experimental design setup (Image generated on 7/6/26 by Grok, xAI)

Standard (Next Generation Science Standards)

MS-LS1-5 Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms.

Grade Level

Middle School

Time Required:

- Setup: 1 class period (45-60 minutes)
- Growth Period: 3 weeks, tracked M/W/F
- Conclusion and Analysis: 1-2 class periods

Lesson Overview:

Students will test the effects of different colors of artificial light on plant growth as they think through the practicalities and challenges of growing plants on the moon.

Learning Objectives:

Students will be able to:

1. Make a hypothesis about the effect of light color on plant growth
2. Create a controlled experiment by manipulating one variable
3. Connect findings to a larger concept of growing plants on the moon

Key Vocabulary:

- Photosynthesis
- Light spectrum
- Germination
- Variable (Independent, dependent, control)

Materials:

- Lettuce seeds, 3-4 per group (or other fast-growing seeds)
- Plastic cups or other uniform growing containers
- Potting soil (one bag per class)
- Colored LED bulbs, grow lights, or light filters
- Graduated cylinder
- Rulers

Experimental Design

Independent Variable

Light color:

- White light (control), red, yellow, green, blue (students choose one color)

Dependent variable

- Plant growth (in centimeters)

Controlled variables

- Distance to lamp
- Watering schedule
- Pot/growth container
- Soil used

Optional Arduino Extension:

The Arduino Uno and SEEED Sensor Kit can be used to track different values throughout the experiment. One suggested application is to use the photoresistor to find light values for each

color used in the experiment. The diagram and example code below can be used to track light intensity and view values in the serial monitor. Testing values for each color of light can help students make informed predictions on the outcomes of different colors of artificial light on plant growth.

Follow pictures to set up equipment, then copy code into Arduino IDE to track values from the sensor.



1. Attach Arduino Uno to the bottom of the Sensor Kit Base. (Ensure that the Uno is pushed in all the way.)



2. Use cord to connect the Uno to a computer. Check that green light is on (as shown in the image) to ensure that the base is powered.

Example code:

```
float lightval;
void setup() {
    Serial.begin(9600);
}
void loop() {
    lightval = analogRead(A3);
```

```
    Serial.print("Light:");
    Serial.println(lightval);

    delay(3000);
}
```

Procedure

Setup (Day 1)

1. Introduce the challenges of growing plants on the moon. What equipment and supplies do astronauts require?
2. Transition to experiment
3. Students separate into groups and complete hypothesis steps on their worksheet.
4. Fill plastic pots with 1.5 cups of soil and 3 seeds per pot.
5. Measure 60 mL of water in a graduated cylinder and water pots.
6. Each group is assigned a color and places their pot 6 inches below the light.

Growth Period (3 weeks)

7. After germination, students should thin their pots by leaving only one seedling per pot. (This ensures that multiple seedlings are not competing for resources)
8. Students track growth of their different plants every Monday, Wednesday, and Friday (or during a modified schedule if necessary) and continue to water their plants (60 mL - or as directed by seed growth instructions - each session). They should record height and qualitative observations on the worksheet packets.
9. Throughout the experiment, ensure that other variables are controlled throughout the space. (Examples: temperature, amount of water, proximity to light)

Conclusion & Analysis (Final Day)

10. Using all student data, calculate class averages for plant height under each light on the board.
11. Discuss findings. Which light color resulted in the most growth? What other qualitative observations were made during the growth period?
12. Relate to light spectrum and optimal growth conditions. How do different colors of light affect growth? What is the best color of artificial light for plant growth based on our findings?
13. Lunar plant growth application: How would astronauts use our findings to grow plants on the moon? What equipment would they need?
14. Optional Arduino extension: how do values found using the Arduino photoresistor support our findings?

Guiding Questions

- What do our growth outcomes tell us about artificial light systems, and how could they present difficulties for astronauts?
- How can scientists overcome the challenges of growing plants on the moon?

Concluding Activity (Claim, Evidence, Reasoning & What If?)

On their worksheets, students complete a short Claim, Evidence, Reasoning paragraph to summarize their findings on the connection between light color and plant growth. Students can also complete the What If? section to expand their understanding beyond the scope of the activity. (What if... you varied light brightness instead of color? You needed to buy equipment to design a greenhouse on the moon?)

Cross-Cutting Concepts:

- Cause and effect relationships
- Patterns